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1.) A 200 MHz left-hand circularly polarized plane wave with an electric field modulus of 5 V/m is normally incident in air on a dielectric medium with ϵ_r = 4 and occupies the region defined by z ≥ 0. (20 points total)

LHC

lossless

(a) Write an expression for the electric field phasor of the incident wave given that the field is a positive maximum at z = 0 and t = 0. (5 points)

$\omega = 2\pi f = 2\pi(200\times10^6) = 4\pi\times10^8$ rad/s

$k_1 = \frac{\omega}{c} = \frac{4\pi\times10^8}{3\times10^8} = \frac{4}{3}\pi$ rad/m

$a_0 = 5$ V/m

$\tilde{E}^i = a_0 \hat{e} e^{-jk_1 z}$

$= a_0(\hat{x} + j\hat{y}) e^{-jk_1 z}$

$= 5(\hat{x} + j\hat{y}) e^{-j\frac{4\pi z}{3}}$

$\boxed{\tilde{E}^i = 5(\hat{x} + j\hat{y}) e^{-j\frac{4\pi}{3}z} \text{ V/m}}$

(b) Calculate the reflection and ~~transmission coefficients~~. (5 points) $\mu_r = 1$, Air, also not given

Medium 1: Air, $\epsilon_{r_1} = 1$, $\mu_{r_1} = 1$

Medium 2 dielectric $\epsilon_r = 4$

$\eta_1 = \sqrt{\frac{\mu_{r_1}}{\epsilon_{r,1}}}\eta_0$

$= \sqrt{\frac{1}{1}}\, 120\pi$

$= 120\pi\ \Omega$

$\eta_2 = \sqrt{\frac{\mu_{r_2}}{\epsilon_{r_2}}}\eta_0$

$= \sqrt{\frac{1}{4}}\, 120\pi$

$= 60\pi\ \Omega$

$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{60\pi - 120\pi}{60\pi + 120\pi} = \boxed{-\frac{1}{3}}$

$\tau = 1 + \Gamma = 1 - \frac{1}{3} = \boxed{\frac{2}{3}}$

(c) Write expressions for the electric field phasors of the reflected wave, the transmitted waves, and the total field in the region $z \le 0$. (5 points)

Medium 1 (air $\varepsilon_{r1} = 1$) $k_1 = \frac{\omega\sqrt{1}}{c} = \frac{4\pi\times10^8}{3\times10^8} = \frac{4}{3}\pi$ rad/m

Medium 2 (dielectric $\varepsilon_{r2} = 4$), $k_2 = \frac{\omega\sqrt{\varepsilon_{r2}}}{c} = \frac{(4\pi\times10^8)\sqrt{4}}{3\times10^8} = \frac{8}{3}\pi$ rad/m

Phasor template (Med 2 $z \ge 0$)

Reflective wave

$$\tilde{E}^r = \Gamma \, a_0 \hat{e} \, e^{+jk_1 z}$$

$$= -\frac{1}{3}(5)(\hat{x} + j\hat{y}) e^{+j\frac{4}{3}\pi z}$$

$$\boxed{\tilde{E}^r = -\frac{5}{3}(\hat{x} + j\hat{y}) e^{+j\frac{4}{3}\pi z} \text{ V/m}}$$

transmitted wave

$$\tilde{E}^t = \tau \cdot a_0 \hat{e} \, e^{-jk_2 z}$$

$$= \frac{2}{3} \cdot 5(\hat{x} + j\hat{y}) e^{-jk_2 z}$$

$$\boxed{\tilde{E}^t = \frac{10}{3}(\hat{x} + j\hat{y}) e^{-j\frac{8}{3}\pi z} \text{ V/m}}$$

Total field in medium 1

$$\tilde{E}_1 = \tilde{E}^i + \tilde{E}^r$$

$$= 5(\hat{x} + j\hat{y}) e^{-j\frac{4\pi}{3} z} + \left(-\frac{5}{3}\right)(\hat{x} + j\hat{y}) e^{+j\frac{4}{3}\pi z}$$

$$\boxed{\tilde{E}_1 = 5(\hat{x} + j\hat{y}) \left[e^{-j\frac{4\pi}{3} z} - \frac{1}{3} e^{+j\frac{4}{3}\pi z} \right] \text{ V/m}}$$

(d) Determine the percentages of the incident average power reflected by the boundary and transmitted into the second medium. (5 points)

$$\% P_r = 100\,|\Gamma|^2 = 100\left|-\frac{1}{3}\right|^2 = \boxed{11.1\%}$$

$$\% P_t = 100\,|\tau|^2 \left(\frac{\eta_1}{\eta_2}\right) = \frac{120\pi}{60\pi} = \boxed{88.9\%}$$

Check work $11.1\% + 88.9\% = 100\%$ Energy conserved

2. Repeat the above problem but replace the dielectric medium with a poor conductor characterized by ϵ_r = 2.25, μ_r = 1, and σ = 10^−4 S/m (20 points total)

(a) Write an expression for the electric field phasor of the incident wave given that the field is a positive maximum at z = 0 and t = 0. (5 points)

Same as P1

$$\boxed{\tilde{E} = 5(\hat{x} + j\hat{y})e^{-j\frac{4}{5}\pi z} \text{ V/m}}$$

(b) Calculate the reflection and transmission coefficients. (5 points)

Classify type of media

$$\frac{\sigma_2}{\omega\varepsilon_2} = \frac{\sigma_2}{\omega \varepsilon_{r2}\varepsilon_0} = \frac{10^{-4}}{(4\pi\times10^8)(2.25)(8.854\times10^{-12})} = 4\times10^{-3}$$

Since $\ll 1$, it's low loss dielectric

$$\eta_2 \approx \sqrt{\frac{\mu}{\varepsilon'}}\left(1 + j\frac{\sigma}{2\omega\varepsilon'}\right) = \frac{\eta_0}{\sqrt{\varepsilon_{r2}}} = \frac{120\pi}{\sqrt{2.25}} = 80\pi\ \Omega$$

Part C

$$\alpha_2 = \frac{\sigma_2\eta_0}{2\sqrt{\varepsilon_{r2}}} = \frac{(10^{-4})(120\pi)}{2\sqrt{2.25}} = 1.26\times10^{-12}\ \text{Np/m}$$

$$\beta_2 = \frac{\omega\sqrt{\varepsilon_{r2}}}{c} = \frac{(4\pi\times10^8)(1.5)}{3\times10^8} = 2\pi\ \text{rad/m}$$

$$\Gamma = \frac{\eta_2-\eta_1}{\eta_2+\eta_1} = \frac{80\pi-120\pi}{80\pi+120\pi} = \boxed{-\frac{1}{5}}$$

$$\tau = 1+\Gamma = 1-\frac{1}{5} = \boxed{\frac{4}{5}}$$

(c) Write expressions for the electric field phasors of the reflected wave, the transmitted waves, and the total field in the region z ≤ 0. (5 points)

$\tilde{E}^r = \Gamma \cdot 5(\hat{x} + j\hat{y}) e^{+j\frac{4\pi}{3}z} = -(\hat{x} + j\hat{y}) e^{+j\frac{4\pi z}{3}}$ V/m

$\tilde{E}^t = \tau \cdot 5(\hat{x} + j\hat{y}) e^{-\alpha_2 z} e^{-j\beta_2 z} = 4(\hat{x} + j\hat{y}) e^{-1.26 \times 10^{-2} z} e^{-j2\pi z}$ V/m

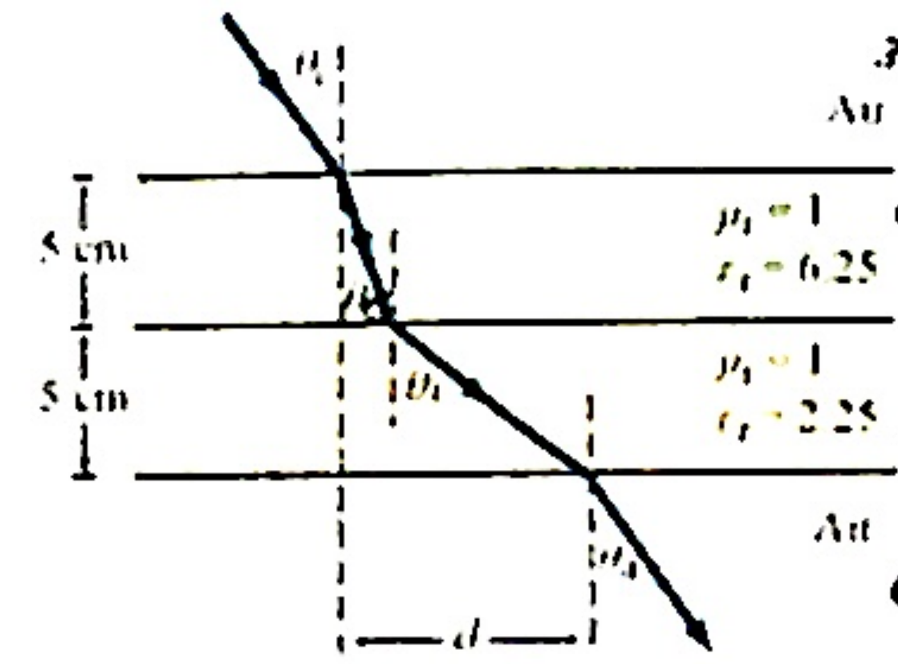
$\boxed{\tilde{E}_1 = \tilde{E}^i + \tilde{E}^r = 5(\hat{x} + j\hat{y})\left(e^{-j\frac{4\pi z}{3}} - \frac{1}{5} e^{j\frac{4\pi z}{3}}\right) \text{V/m}}$

(d) Determine the percentages of the incident average power reflected by the boundary and transmitted into the second medium. (5 points)

$\% P_r = 100|\Gamma|^2 = 100(-\frac{1}{5})^2 = \boxed{4\%}$

$\% P_t = 100|\tau|^2 \frac{\eta_1}{\eta_2} = 100\left|\frac{4}{5}\right|^2 \frac{120\pi}{80\pi} = \boxed{96\%}$

3. A parallel-polarized plane wave is incident from air at an angle θ_i = 30∘ onto a pair of dielectric layers, as shown above. (10 points total)



$\varepsilon_{r1} = 1$ b/c air

$\varepsilon_{r2} = 6.25$

$\varepsilon_{r3} = 2.25$

$\varepsilon_r = 1$

(a) Determine the angles of transmission θ_2, θ_3, and θ_4. (5 points)

Snell's law

Stage 1 Air $\theta_i = 30°$

Stage 2

$\sin(\theta_2) = \sin\theta_1 \sqrt{\frac{\varepsilon_{r1}}{\varepsilon_{r2}}}$

$\sin(\theta_2) = \sin 30° \sqrt{\frac{1}{6.25}}$

$\sin(\theta_2) = 0.2$

$\boxed{\theta_2 = 11.54°}$

Stage 3

$\sin(\theta_3) = \sin(\theta_2) \sqrt{\frac{\varepsilon_{r2}}{\varepsilon_{r3}}}$

$\sin(\theta_3) = \sin(11.54°) \sqrt{\frac{6.25}{2.25}}$

$\boxed{\theta_3 = 19.48°}$

Stage 4

$\sin(\theta_4) = \sin(\theta_3) \sqrt{\frac{\varepsilon_{r3}}{\varepsilon_{r4}}}$

$\theta_4 = \sin^{-1}\left(\sin(19.48°) \sqrt{\frac{2.25}{1}}\right)$

$\boxed{\theta_4 \approx 30.01°}$

(b) Determine the lateral distance d. (5 points)

$d = t_1 \tan\theta_2 + t_2 \tan\theta_3$ $\quad t_2 = t_1 = 0.05\,m$

$= 0.05 \tan(11.54) + 0.05 \tan(19.48°)$

$\boxed{d = 0.03\,m}$

SOH CAH TOA

$\tan(\theta) = \frac{O}{A}$

$\tan(\theta) = \frac{d_1}{t_1}$

$d_1 = t$

4. A $\boxed{\text{TE}}$ wave propagating in a dielectricfilled waveguide of unknown permittivity has dimensions a = 5 cm and b = 3 cm. If the x componentof its electric field is given by

$=0.05$ $\quad$ 0.03

$$E_x = -36\cos(40\pi x)\sin(100\pi y)\sin\left(\left(\underbrace{2.4\pi \times 10^{10}}_{\omega =}\right)t - 52.9\pi z\right)\ (\text{V/m}),$$

$\frac{m\pi}{a}x$ $\quad$ $\frac{n\pi y}{b}$

determine: (20 points total)

(a) the mode number, (5 points)

$E_x = \cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)\sin(\omega t - \beta z)$

$\frac{m\cancel{\pi}\cancel{x}}{a} = 40\cancel{\pi}\cancel{x}$ $\quad$ $\frac{n\cancel{\pi}\cancel{y}}{b} = 100\cancel{\pi}\cancel{y}$

$m = 2$ $\quad$ $n = 3$

$TE_{mn} = \boxed{TE_{23}}$

(b) ϵ_r of the material in the guide, (5 points) $\omega = 2.4\pi \times 10^{10}$

$\varepsilon_r = \frac{c^2}{\omega^2}\left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right]$

$= \frac{(3\times10^8)^2}{(2.4\pi\times10^{10})^2}\left[(52.9\pi)^2 + (40\pi)^2 + (100\pi)^2\right]$

$\boxed{\varepsilon_r = 2.25}$

(c) the cutoff frequency, (5 points)

$$f_{mn} = \frac{u_{p0}}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}\ Hz$$

$$u_{p0} = \frac{c}{\sqrt{\varepsilon_r}} = \frac{3\times10^8}{\sqrt{2.25}} = 200\times10^6\ m/s$$

$$f_{23} = \frac{200\times10^6}{2}\sqrt{40^2 + 100^2}$$

$$\boxed{f_{23} = 10.77\ GHz}$$

(d) and the expression for H_y. (5 points)

$f_{23c} = 10.77\ GHz$

$$f = \frac{\omega}{2\pi} = \frac{2.4\pi\times10^{10}}{2\pi} \Rightarrow 12.00\ GHz$$

$$\eta = \frac{377}{\sqrt{2.25}} = 251.33$$

$$Z_{TE} = \frac{\eta}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}}$$

$$= \frac{251.33}{\sqrt{1-\left(\frac{10.77\ GHz}{12.00\ GHz}\right)^2}}$$

$$= 569.84\ \Omega$$

$$H_y = \frac{E_x}{Z_{TE}} = \frac{-36}{569.84}\cos(40\pi x)\sin(100\pi y)\sin((2.4\pi\times10^{10})t - 52.9\pi z)\ V/m$$

$$\boxed{H_y = -63.17\times10^{-3}\cos(40\pi x)\sin(100\pi y)\sin[(2.4\pi\times10^{10}t) - 52.9\pi z)]\ V/m}$$

5. A narrow rectangular pulse superimposed on a carrier with a frequency of 9.5 GHz was used to excite all possible modes in a hollow guide ($\epsilon_r = 1$) with a = 3 cm and b = 2 cm. If the guide is 100 m in length, how long will it take each of the excited modes to arrive at the receiving end? (8 points)

$a = 0.03 \qquad b = 0.02$

$u_{po} = \frac{c}{\sqrt{\epsilon_r}} = 3\times10^8 \text{ m/s} \qquad u_g = u_{po}\sqrt{1-\left(\frac{f_{mn}}{f}\right)^2}$

Mode	f_c	u_g	$t = L/u_g$
TE_{10}	$f_{10} = \frac{c}{2a} = \frac{3\times10^8}{2(0.03)} = 5\text{ GHz}$ ✓	$3\times10^8\sqrt{1-\left(\frac{5\text{GHz}}{9.5\text{GHz}}\right)^2} = 255.1\times10^6$ m/s	$\frac{100}{255.1\times10^6} = \boxed{0.392\mu s}$
TE_{01}	$f_{01} = \frac{c}{2b} = \frac{3\times10^8}{2(0.02)} = 7.5\text{GHz}$ ✓	$= 3\times10^8\sqrt{1-\left(\frac{7.5\text{GHz}}{9.5\text{GHz}}\right)^2} = 184.14\times10^6$ m/s	$\frac{100}{184.14\times10^6} = \boxed{543.07\text{ ns}}$
TE_{20}	$\frac{u_{po}}{a} = \frac{3\times10^8}{0.03} = 10\text{ GHz}$ ✗ 9.5G < 10G, not excited	D/C, not excited	D/C
TE_{11}, TM_{11}	$\frac{u_{po}}{2}\sqrt{\frac{1}{a^2}+\frac{1}{b^2}} = \frac{3\times10^8}{2}\sqrt{\frac{1}{0.03^2}+\frac{1}{0.02^2}} = 9.01\text{ GHz}$ ✓✓	$u_{po}\sqrt{1-\left(\frac{9.01G}{9.5G}\right)^2} = 95.10\times10^6$ m/s	$\frac{100}{95.10\times10^6} = 1.02\times10^{-6} = \boxed{1.02\mu s}$